

Dataset	Sample (%)	Sample size	Random MAPE (%)	Diversity MAPE (%)	Best mean-K K-means	Best overall
Apache	10	19	17.88	14.33	K-means Hamming (19.43%)	Diversity
Apache	20	38	16.26	15.78	K-means Hamming (16.62%)	Diversity
Apache	30	58	15.82	14.73	K-means default (15.43%)	Diversity
Apache	40	77	15.62	14.91	K-means default (15.24%)	Diversity
Apache	50	96	15.53	13.75	K-means default (15.13%)	Diversity
Apache	60	115	15.54	14.21	K-means default (15.07%)	Diversity
Apache	70	134	15.69	13.73	K-means default (14.98%)	Diversity
BDBC	10	256	423.03	392.93	K-means default (198.95%)	K-means default (mean K)
BDBC	20	512	435.33	514.12	K-means default (265.70%)	K-means default (mean K)
BDBC	30	768	431.93	425.01	K-means default (300.87%)	K-means default (mean K)
BDBC	40	1024	430.25	432.75	K-means default (326.31%)	K-means default (mean K)
BDBC	50	1280	430.06	479.54	K-means default (348.94%)	K-means default (mean K)
BDBC	60	1536	435.82	565.79	K-means default (364.96%)	K-means default (mean K)
BDBC	70	1792	423.42	700.57	K-means default (377.19%)	K-means default (mean K)
BDBJ	10	18	45.07	47.17	K-means default (39.53%)	K-means default (mean K)
BDBJ	20	36	40.21	32.07	K-means default (39.85%)	Diversity
BDBJ	30	54	39.27	31.94	K-means Hamming (36.67%)	Diversity
BDBJ	40	72	38.04	33.83	K-means Hamming (36.29%)	Diversity
BDBJ	50	90	38.38	33.75	K-means Hamming (36.45%)	Diversity
BDBJ	60	108	37.72	33.15	K-means Hamming (36.78%)	Diversity
BDBJ	70	126	37.94	35.04	K-means Hamming (37.31%)	Diversity
LLVM	10	102	2.54	2.34	K-means Hamming (2.53%)	Diversity
LLVM	20	205	2.51	2.49	K-means Hamming (2.43%)	K-means Hamming (mean K)
LLVM	30	307	2.44	2.28	K-means Hamming (2.41%)	Diversity
LLVM	40	410	2.41	2.46	K-means Hamming (2.40%)	K-means Hamming (mean K)
LLVM	50	512	2.43	2.69	K-means Hamming (2.40%)	K-means Hamming (mean K)
LLVM	60	614	2.4	2.99	K-means Hamming (2.39%)	K-means Hamming (mean K)
LLVM	70	717	2.41	2.7	K-means default (2.30%)	K-means default (mean K)